

GIRISH VUYYURU

AI Solutions Architect • Generative AI & Agentic Systems

Houston, TX • (346) 434-8296 • girishsap45@gmail.com
gireeshvuyyuru501-design.github.io/portfolio • linkedin.com/in/girish-genai-engineer • github.com/gireeshvuyyuru501-design
STEM OPT EAD — no sponsorship required • Open to remote or hybrid

SUMMARY

Generative AI engineer with end-to-end design ownership of production agentic and retrieval systems. I set the architecture — agent topology, retrieval strategy, model routing, evaluation gates — against hard non-functional requirements: sub-150ms at 5,000+ daily requests, measurable hallucination ceilings, and a defined inference budget. Three years across Google, Tesla, and Deloitte; nine production systems shipped.

SELECTED SYSTEMS

Enterprise AI Orchestration Platform — Deloitte, 2026

LangGraph supervisor routing to 6 specialist agents over a FastMCP server (9 tools, A2A), with RAG across 8 vector databases on AWS, Azure, and GCP. Chose a supervisor pattern over a peer mesh for auditable routing and clean failure isolation. 500+ daily users at sub-150ms; 43% hallucination reduction; \$2,800/month saved.

LangGraph • FastMCP • AWS Bedrock • Azure AI • LangSmith • RAGAS

Enterprise MCP Knowledge Assistant

Production MCP server (9 tools) connecting Claude and GitHub Copilot to enterprise knowledge, secured with Azure Entra ID SSO and RBAC, evaluated continuously in LangSmith and RAGAS.

FastMCP • Claude • pgvector github.com/gireeshvuyyuru501-design/mcp_employee_knowledge_assistant

AI Incident Triage Copilot (AIOps)

Runbook retrieval via TF-IDF and cosine similarity, with severity classification and root cause analysis behind a FastAPI service.

AIOps • NLP • FastAPI github.com/gireeshvuyyuru501-design/ai-incident-triage-copilot

Customer Churn MLOps Pipeline

End-to-end pipeline over 7,000+ records at 79.68% recall, provisioned with Terraform, shipped through GitHub Actions, with drift monitoring in production.

MLflow • scikit-learn • Terraform github.com/gireeshvuyyuru501-design/customer-churn-mlops-pipeline

EXPERIENCE

Deloitte Senior Generative AI & Agentic AI Engineer	Oct 2025 – Present Houston, TX
– Design authority across the client GenAI portfolio — agent topology, retrieval strategy, and evaluation gates — for Fortune 500 engagements. – Designed the inference cost model — model routing, prompt versioning, token optimization in LangSmith — reducing LLMOps spend \$2,800/month. – Owned deployment architecture on Docker, Kubernetes, and CI/CD — 9 production systems delivered as lead engineer, 60% fewer defects.	
Tesla AI/ML Engineer	Aug 2024 – Sep 2025 Miami, FL
– Designed a LangGraph and CrewAI multi-agent system for document processing and workflow automation, removing 55% of manual effort — owned the design through production deployment. – Set the observability standard for LLM services at 94% coverage in LangSmith and halved release cycles through CI/CD design; improved NLP accuracy 28% via LoRA/QLoRA fine-tuning, serving 10,000+ daily requests at sub-150ms latency.	
Google AI/ML Engineer	Jun 2023 – Jul 2024 Mountain View, CA
– Designed a production AI platform on LangGraph, LangChain, and RAG (pgvector, FAISS, Pinecone) serving 500+ daily users — 43% hallucination reduction and 65% latency improvement. – Designed the monitoring and evaluation approach across AWS Lambda and GCP Vertex AI (35% fewer production incidents), and architected backend APIs for AI orchestration, semantic search, and document processing.	

TECHNICAL ARCHITECTURE

Agentic LangGraph, LangChain, FastMCP / MCP, CrewAI, LlamaIndex, A2A patterns, tool-server design
Retrieval pgvector, FAISS, Pinecone, Weaviate, ChromaDB, Azure AI Search, hybrid search, reranking, chunking
Models Claude API, GPT-4o, AWS Bedrock, Azure OpenAI, GCP Vertex AI, LLaMA 3, Mistral, LoRA / PEFT / QLoRA
Evaluation RAGAS, LangSmith, DeepEval, MLflow, guardrails, drift detection
Platform Python, FastAPI, PostgreSQL, Redis, Docker, Kubernetes, GitHub Actions, AWS, Azure, GCP

EDUCATION & CERTIFICATIONS

Education M.S. Computer Science, Texas A&M University (4.0 GPA) • B.E. Computer Engineering, Vignan's University
Certifications 16 Anthropic (Claude Code, MCP Advanced, Agent Skills, Subagents, Bedrock, Vertex AI) • AWS ML Specialty